$T_0 < 1ps$，需包括高阶非线性.

$$\frac{\partial U}{\partial \xi} + i\frac{sgn(\beta_2)}{2L_D}\frac{\partial^2 U}{\partial \tau^2} = \frac{sgn(\beta_3)}{6L_D'}\frac{\partial^3 U}{\partial \tau^3} + i\frac{e^{-\alpha\xi}}{L_{NL}}\left(|U|^2U + is\frac{\partial}{\partial \tau}(|U|^2U) - \tau_R U\frac{\partial |U|^2}{\partial \tau}\right)$$

- $L_D = \frac{T_0^2}{|\beta_2|}$，
- $L_D' = \frac{T_0^3}{|\beta_3|}$，
- $L_{NL} = \frac{1}{\gamma P_0}$
- s 与 τ_R 分别描述了自变陡与脉冲内拉曼散射.

$$s = \frac{1}{\omega_0 T_0}, \quad \tau_R = \frac{T_R}{T_0}$$

自陡峭.

来源: 群速度对光强的依赖.

考虑无色散情形. $\beta_2 = \beta_3 = 0$

Def 归一化距离 $\xi = z/L_{NL}$

$$\frac{\partial u}{\partial \xi} + s\frac{\partial}{\partial \tau}(|u|^2 u) = i|u|^2 u$$

由 $u = \sqrt{I}\exp(i\phi)$

$$\begin{cases} \frac{\partial I}{\partial \xi} + 3sI\frac{\partial I}{\partial \tau} = 0 \\ \frac{\partial \phi}{\partial \xi} + sI\frac{\partial \phi}{\partial \tau} = 0 \end{cases}$$

由特征选择方法, 通解:

$$I(\xi, \tau) = f(\tau - 3sI\xi) \qquad (4.4.7)$$

初始条件:

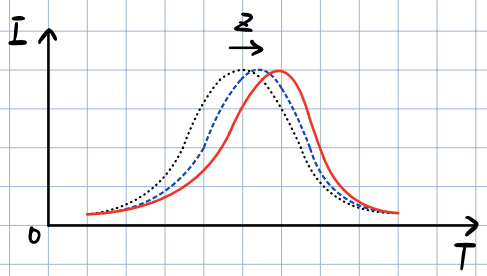
$$I(0, \tau) = f(\tau) \qquad f \text{描述了} z=0 \text{脉冲形状}$$

- 每个 τ 从其初始位置沿直线移动, 直线斜率与光强相关.

  $\longrightarrow$ 脉冲畸变.

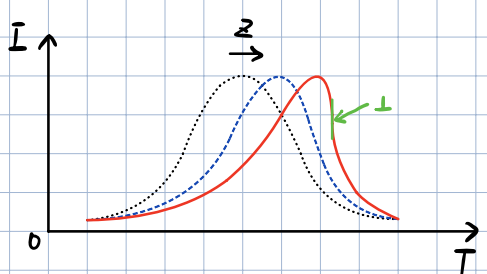
如: $I(0, \tau) = f(\tau) = e^{-\tau^2}$. (Gaussian pulse).

$$I(\xi, \tau) = e^{-(\tau - 3sI\xi)^2}$$



- 随脉冲传输，峰值移向脉冲后沿，脉冲变得不对称．后沿随传输越来越陡．　— 自陡陡 Self-steepen
- 实质：群速度强度相关，峰的移动速度比两翼快．

自陡陡最后会形成光波冲击 optical shock．
前沿某处的斜率越来越陡达 90°．



令 $\frac{\partial T}{\partial z} \to \infty$．有

$$z_s = \sqrt{\frac{e}{2}} \frac{L_{NL}}{35} \approx 0.39 \frac{L_{NL}}{5}.$$

- 自陡陡也影响 SPM 频谱展宽．
  - 红移峰比蓝移峰更强．
  - 蓝端比红端有更宽的 SPM 展宽量．

- pulse 形状不对称 ⟶ 展宽频谱不对称.

  SPM 中, pulse 后沿 ⟶ 蓝移分量.

  后沿更陡 ⟶ 蓝移端更宽.

  能量分布在更宽的范围 ⟶ 强度↓.